

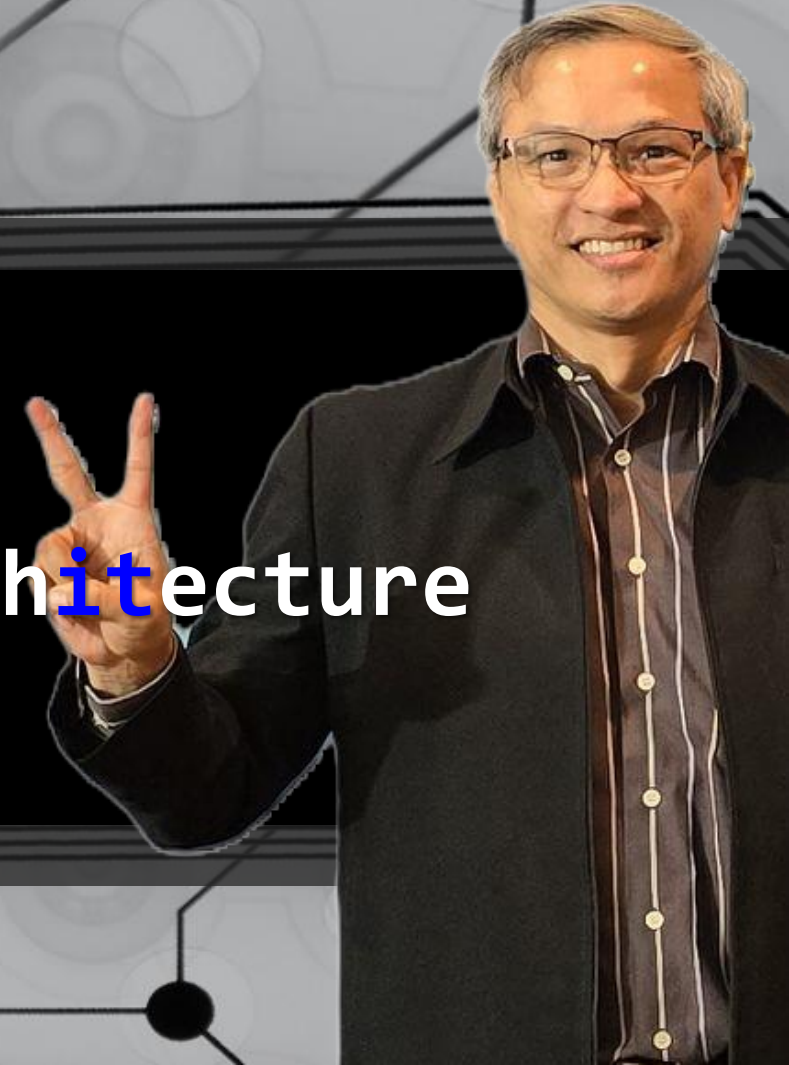


COS2108 – Chapter 2

โครงสร้างและสถาปัตยกรรมคอมพิวเตอร์

Computer Organization and Architecture

ผศ.ชาตรี นิลน้ำเพชร



หัวข้อบรรยาย - Topic

- Digital Logic
 - Boolean Algebra
 - Logic Gates and Operations
 - Combinational Circuits
 - Boolean Functions
 - Multiplexer / Demultiplexer
 - Decoder / Encoder
 - Adders
 - Sequential Circuits
 - Flip-Flops

Boolean Algebra

Name	AND form	OR form
Identity law	$1A = A$	$0 + A = A$
Null law	$0A = 0$	$1 + A = 1$
Idempotent law	$AA = A$	$A + A = A$
Inverse law	$A\bar{A} = 0$	$A + \bar{A} = 1$
Commutative law	$AB = BA$	$A + B = B + A$
Associative law	$(AB)C = A(BC)$	$(A + B) + C = A + (B + C)$
Distributive law	$A + BC = (A + B)(A + C)$	$A(B + C) = AB + AC$
Absorption law	$A(A + B) = A$	$A + AB = A$
De Morgan's law	$\overline{AB} = \bar{A} + \bar{B}$	$\overline{A + B} = \bar{A}\bar{B}$

Example

$$A + \bar{A}B = A + B$$

$$A(\bar{A} + B) = AB$$

$$\overline{A.B.C} = \bar{A} + \bar{B} + \bar{C}$$

$$\overline{A+B+C} = \bar{A}.\bar{B}.\bar{C}$$

$$\overline{A.B.C.D} = \bar{A} + \bar{B} + \bar{C} + \bar{D}$$

$$\overline{A+B+C+D} = \bar{A}.\bar{B}.\bar{C}.\bar{D}$$



Boolean Algebra Simplification

- $A.(A + B)$

	$A.(A+B)$	Start
multiply:	$AA + AB$	Distributive Law
but:	$AA = A$	Idempotent Law
then:	$A + AB$	Reduction
thus:	$A(1 + B)$	Annulment Law
equals to:	A	Absorption Law

Boolean Algebra Simplification

- $(A + B).(A + C)$

	$(A + B).(A + C)$	Start
multiply:	$AA + AC + AB + BC$	Distributive law
but:	$AA = A$	Idempotent Law
then:	$A + AC + AB + BC$	Reduction
however:	$A + AC = A$	Absorption Law
thus:	$A + AB + BC$	Distributive Law
again:	$A + AB = A$	Absorption Law
thus:	$A + BC$	Result

Boolean Algebra Simplification

- $AB.(\bar{B}C + AC)$

	$AB.(\bar{B}C + AC)$	Start
multiply	$AB\bar{B}C + ABAC$	Distributive Law
again:	$AA = A$	Idempotent Law
then:	$AB\bar{B}C + ABC$	Reduction
but:	$B\bar{B} = 0$	Complement Law
so:	$A.0.C + ABC$	Reduction
becomes:	$0 + ABC$	Reduction
as:	$0 + ABC = ABC$	Identity Law
thus:	ABC	Result

Boolean Algebra Simplification

$$(A + \bar{B} + \bar{C})(A + \bar{B} + C)(A + B + \bar{C})$$

$$A.(A + \bar{B} + C)(A + B + \bar{C}) + \bar{B}.(A + \bar{B} + C)(A + B + \bar{C}) + \bar{C}.(A + \bar{B} + C)(A + B + \bar{C})$$

$$(AA + A\bar{B} + AC)(AA + AB + A\bar{C}) + (A\bar{B} + \bar{B}\bar{B} + \bar{B}C)(A\bar{B} + B\bar{B} + \bar{B}\bar{C}) + (A\bar{C} + \bar{B}\bar{C} + C\bar{C})(A\bar{C} + B\bar{C} + \bar{C}\bar{C})$$

$$AA = A, \bar{B}\bar{B} = \bar{B}, \bar{C}\bar{C} = \bar{C} \text{ (Idempotent Law)}$$

$$B\bar{B} = C\bar{C} = 0 \text{ (Complement Law)}$$

$$(A + A\bar{B} + AC)(A + AB + A\bar{C}) + (A\bar{B} + \bar{B} + \bar{B}C)(A\bar{B} + 0 + \bar{B}\bar{C}) + (A\bar{C} + \bar{B}\bar{C} + 0)(A\bar{C} + B\bar{C} + \bar{C})$$

$$A + AB = A, A + A\bar{B} = A, A + AC = A, A + A\bar{C} = A, \bar{B} + A\bar{B} = \bar{B}, \bar{B} + \bar{B}C = \bar{B}, \bar{C} + A\bar{C} = \bar{C}, \bar{C} + B\bar{C} = \bar{C} \text{ (Absorption Law)}$$

$$A + 0 = A \text{ (Identity Law)}$$

$$(A)(A) + (\bar{B})(A\bar{B} + \bar{B}\bar{C}) + (A\bar{C} + \bar{B}\bar{C})(\bar{C})$$

$$A + (A\bar{B}\bar{B} + \bar{B}\bar{B}\bar{C}) + (A\bar{C}\bar{C} + \bar{B}\bar{C}\bar{C})$$

$$A + A\bar{B} + \bar{B}\bar{C} + A\bar{C} + \bar{B}\bar{C}$$

$$A + \bar{B}\bar{C}$$

Logic Gates and Operations

- แต่ละบิตของข้อมูลจะมีเพียงสองค่าคือ 0 กับ 1
- ในการดำเนินการกับบิตของข้อมูลระหว่างตัวเลข 2 ชุด จะใช้ตารางความจริง (Truth Table) ในการนิยามผลลัพธ์ที่เกิดขึ้น

Operation: $A + B$

- Operator: +
- Operand: A, B

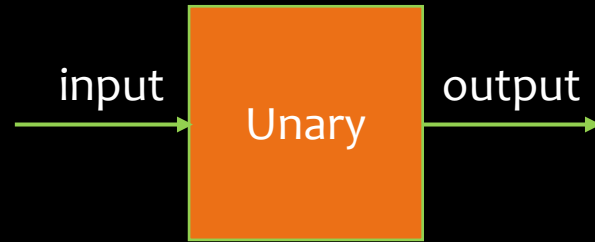
Truth Table

x	y	$\bar{x}$	$x+y$	xy	$x \rightarrow y$	$x \oplus y$	$x \equiv y$
0	0	1	0	0	1	0	1
0	1	1	1	0	1	1	0
1	0	0	1	0	0	1	0
1	1	0	1	1	1	0	1

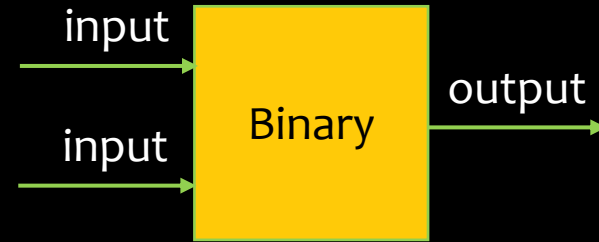
P Q $\neg P$ $P \vee Q$ $P \wedge Q$ $P \rightarrow Q$ $P \oplus Q$ $P \leftrightarrow Q$

Operations and Truth Table

1 – True
0 – False



Unary Operator



Binary Operator

NOT

A	Y
0	1
1	0

AND

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

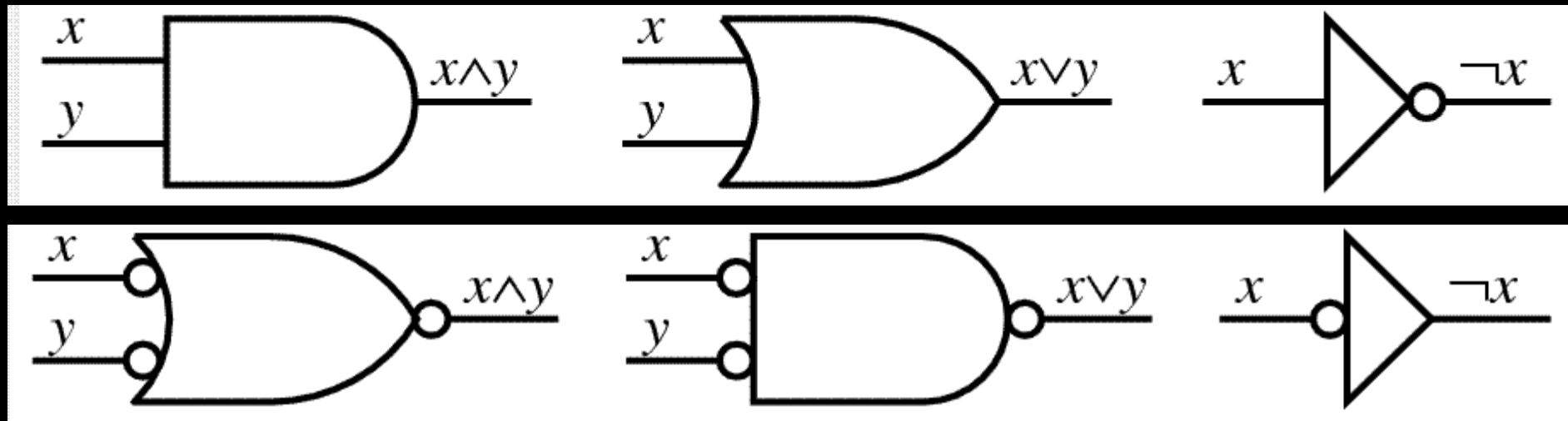
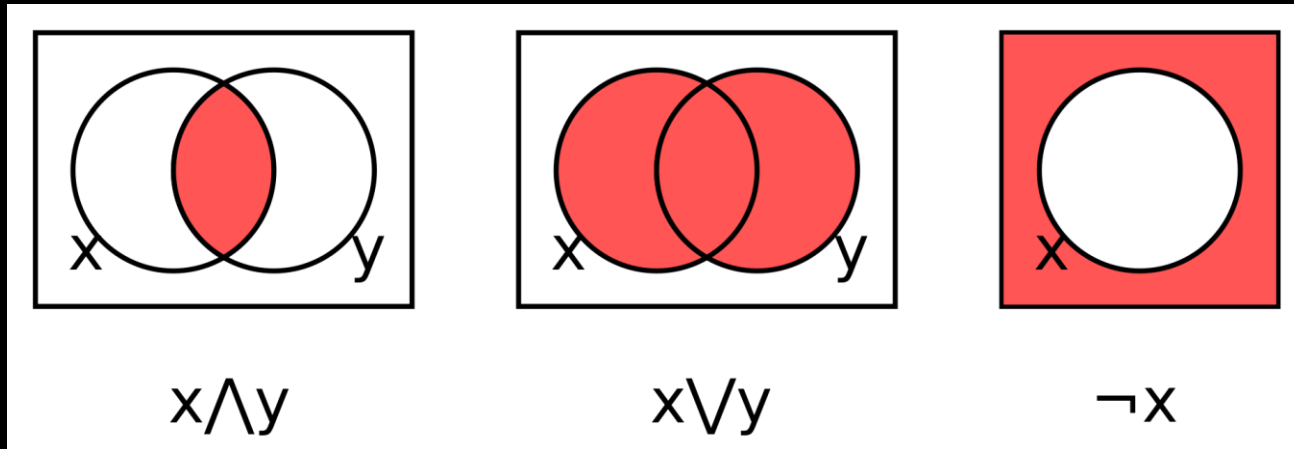
OR

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

XOR

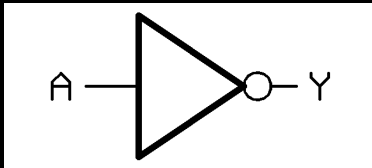
A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

Venn Diagrams VS Logic Gates



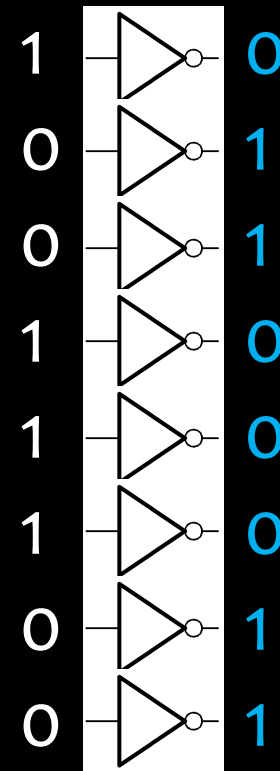
Bit & Byte level operations

NOT Operator



A	Y
0	1
1	0

input output

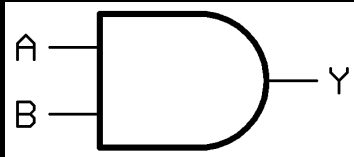


A = 10011100

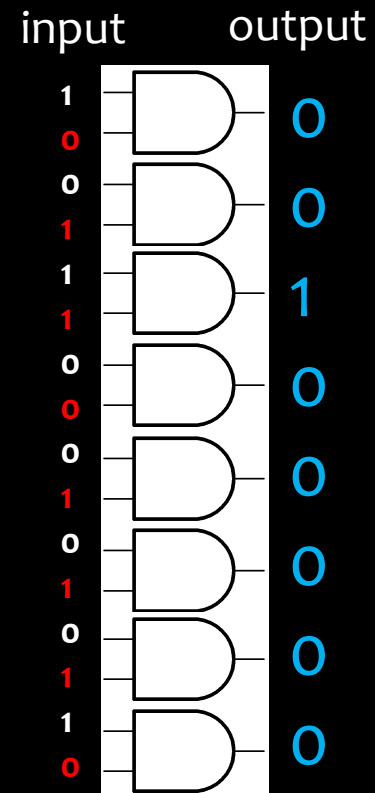
Y = 01100011

Bit & Byte level operations

AND Operator



A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1



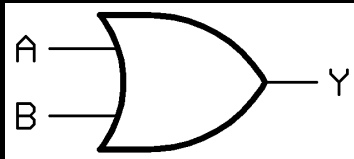
A = 10100001
B = 01101110
Y = 00100000

For A = 0 or 1

- A AND 0 = 0
- 0 AND A = 0

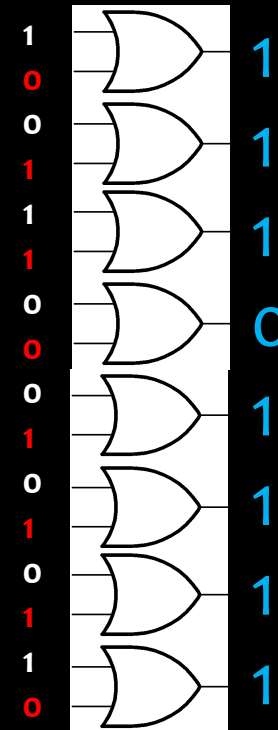
Bit & Byte level operations

OR Operator



A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

input output



A = 10100001

B = 01101110

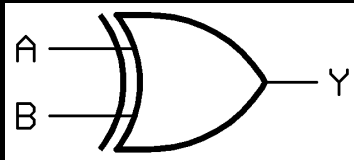
Y = 11101111

For A = 0 or 1

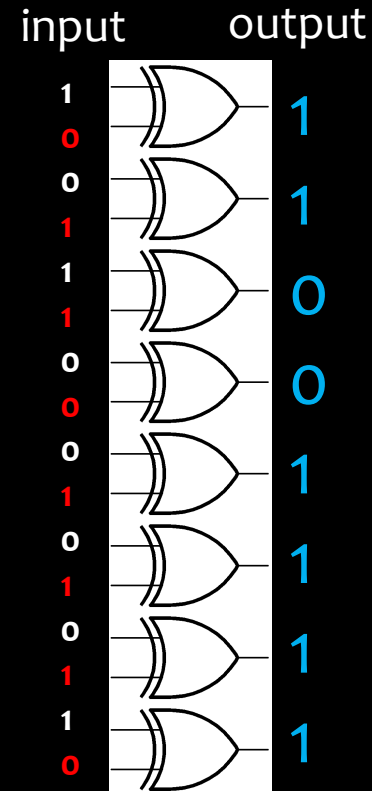
- A OR 1 = 1
- 1 OR A = 1

Bit & Byte level operations

XOR Operator



A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0



A = 10100001

B = 01101110

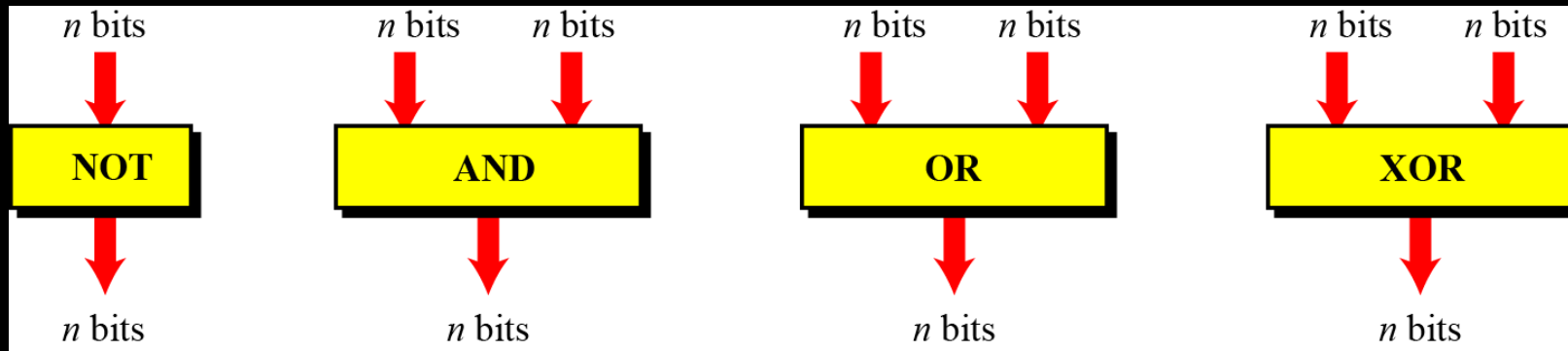
Y = 11001111

$A \text{ XOR } B \leftrightarrow [A \text{ AND } (\text{NOT } B)] \text{ OR } [(\text{NOT } A) \text{ AND } B]$

For A = 0 or 1

- A XOR 1 = NOT A
- 1 XOR A = NOT A

Bit & Byte level operations



NOT	1	0	0	1	1	0	0	0	Input
	0	1	1	0	0	1	1	1	Output

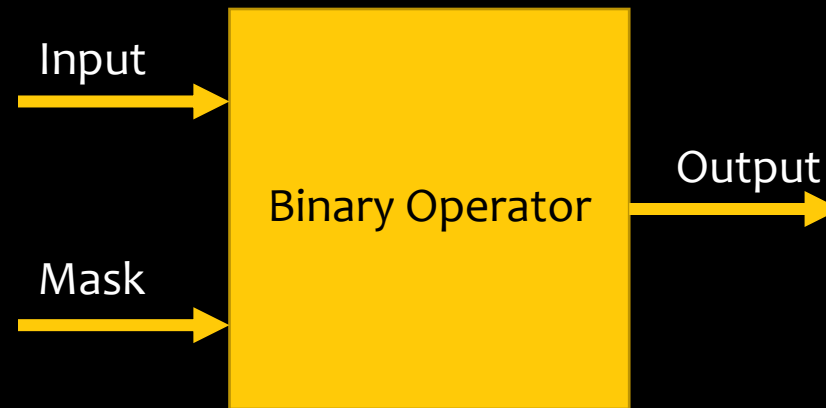
	1	0	0	1	1	0	0	1	Input 1
OR	0	0	1	0	1	1	1	0	Input 2
	1	0	1	1	1	1	1	1	Output

	1	0	0	1	1	0	0	0	Input 1
AND	0	0	1	0	1	0	1	0	Input 2
	0	0	0	0	1	0	0	0	Output

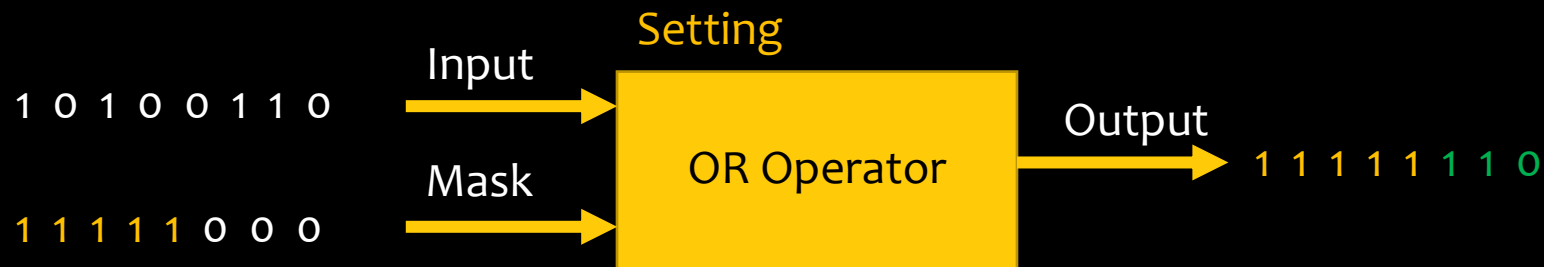
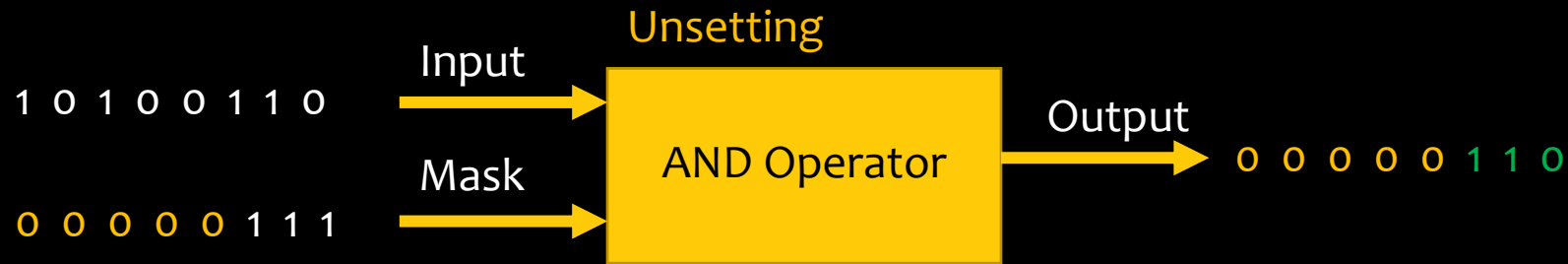
	1	0	0	1	1	0	0	1	Input 1
XOR	0	0	1	0	1	1	1	0	Input 2
	1	0	1	1	0	1	1	1	Output

Applications

- Unary Operator
 - Complementing (NOT)
- Binary Operator
 - Unsetting (AND)
 - Setting (OR)
 - Flipping (XOR)

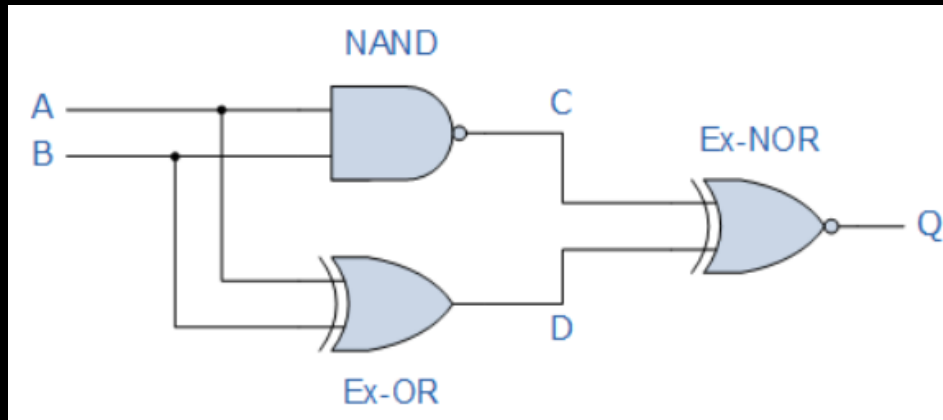


Binary operator application



Boolean Algebra with Truth Table

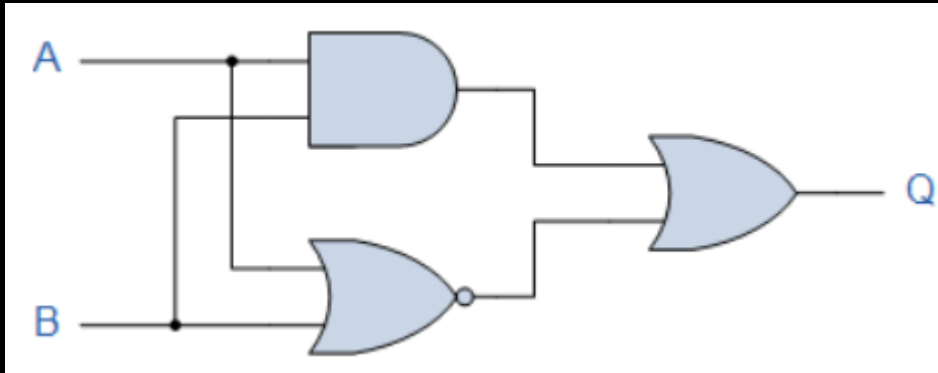
$$Q = \overline{\overline{A}\overline{B}} \oplus (A \oplus B) = A + B$$



Inputs		Output at		
A	B	C	D	Q
0	0	1	0	0
0	1	1	1	1
1	0	1	1	1
1	1	0	0	1

Boolean Algebra with Truth Table

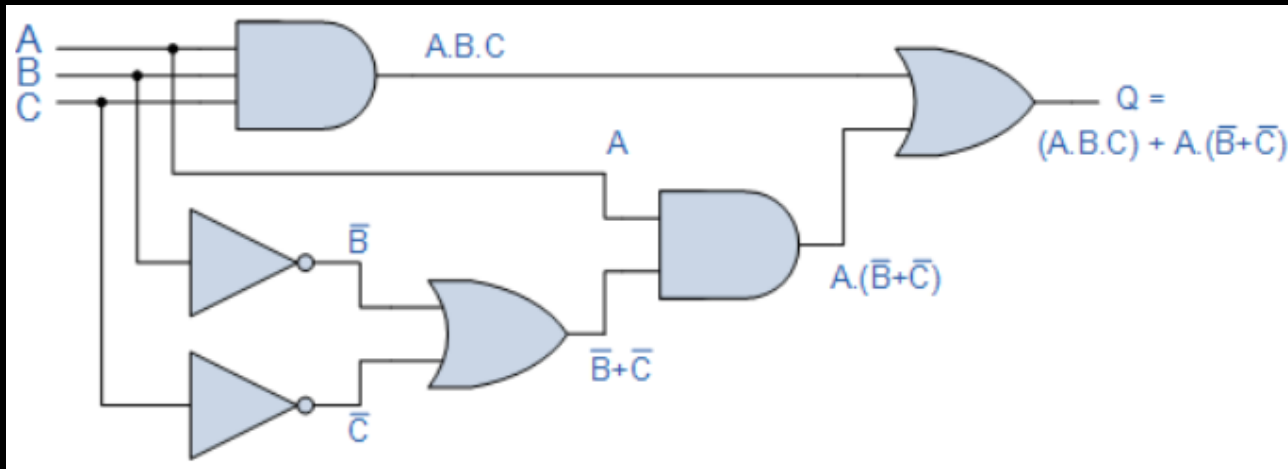
$$Q = AB + \overline{A + B} = \overline{A \oplus B}$$



Inputs		Intermediates		Output
B	A	A.B	$\overline{A + B}$	Q
0	0	0	1	1
0	1	0	0	0
1	0	0	0	0
1	1	1	0	1

Boolean Algebra with Truth Table

$$Q = ABC + A(\bar{B} + \bar{C})$$

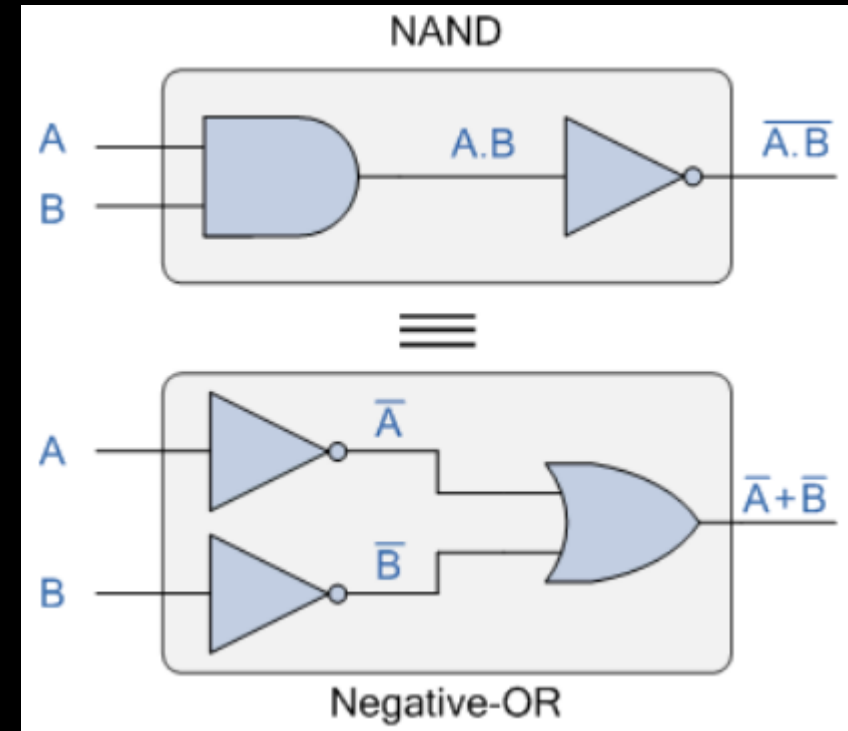


Inputs			Intermediates					Output
C	B	A	A.B.C	$\bar{B}$	$\bar{C}$	$\bar{B} + \bar{C}$	$A(\bar{B} + \bar{C})$	Q
0	0	0	0	1	1	1	0	0
0	0	1	0	1	1	1	1	1
0	1	0	0	0	1	1	0	0
0	1	1	0	0	1	1	1	1
1	0	0	0	1	0	1	0	0
1	0	1	0	1	0	1	1	1
1	1	0	0	0	0	0	0	0
1	1	1	1	0	0	0	0	1

DeMorgan's Law

$$\overline{A \cdot B} = \overline{A} + \overline{B}$$

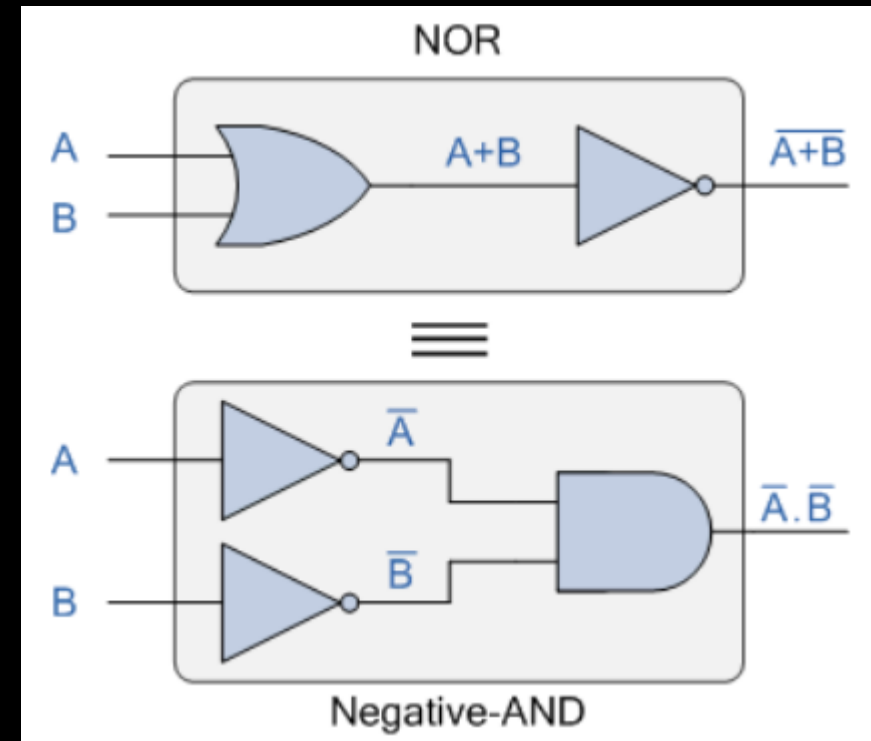
Inputs		Truth Table Outputs For Each Term				
B	A	A.B	$\overline{A \cdot B}$	$\overline{A}$	$\overline{B}$	$\overline{A} + \overline{B}$
0	0	0	1	1	1	1
0	1	0	1	0	1	1
1	0	0	1	1	0	1
1	1	1	0	0	0	0



DeMorgan's Law

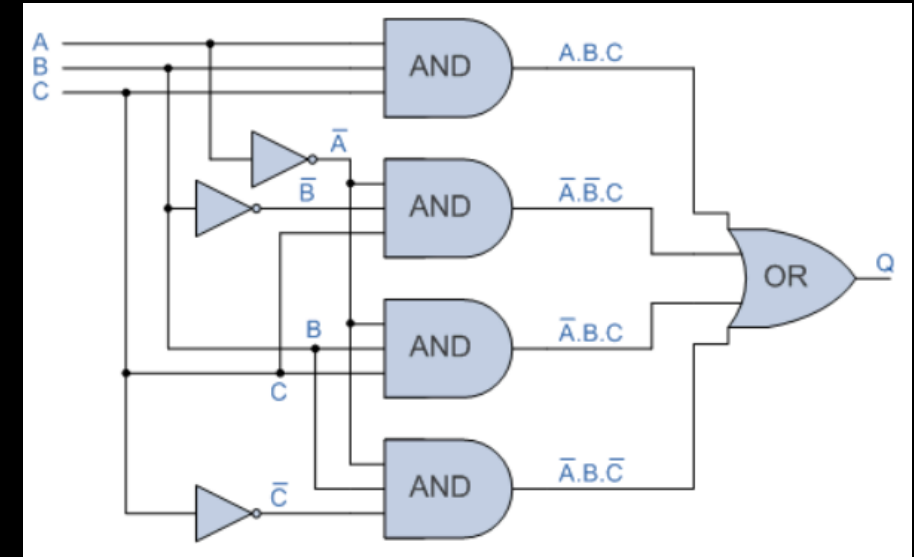
$$\overline{A+B} = \overline{A} \cdot \overline{B}$$

Inputs		Truth Table Outputs For Each Term				
B	A	A+B	$\overline{A+B}$	$\overline{A}$	$\overline{B}$	$\overline{A} \cdot \overline{B}$
0	0	0	1	1	1	1
0	1	1	0	0	1	0
1	0	1	0	1	0	0
1	1	1	0	0	0	0



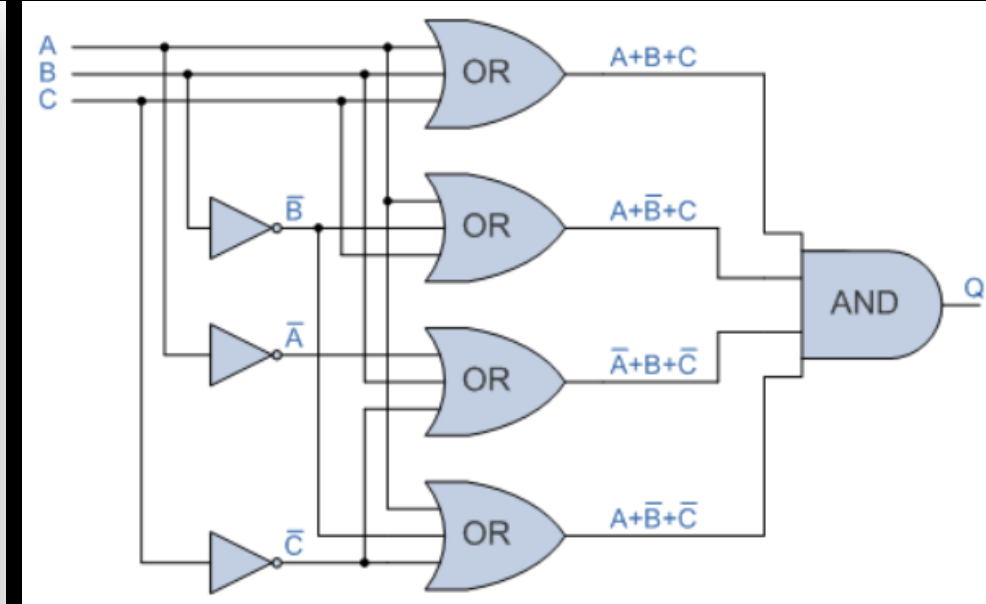
Sum of Product (Min Term)

Inputs			Output	Product
C	B	A	Q	
0	0	0	0	
0	0	1	0	
0	1	0	1	$\bar{A}B\bar{C}$
0	1	1	0	
1	0	0	1	$\bar{A}\bar{B}C$
1	0	1	0	
1	1	0	1	$\bar{A}B.C$
1	1	1	1	$A.B.C$

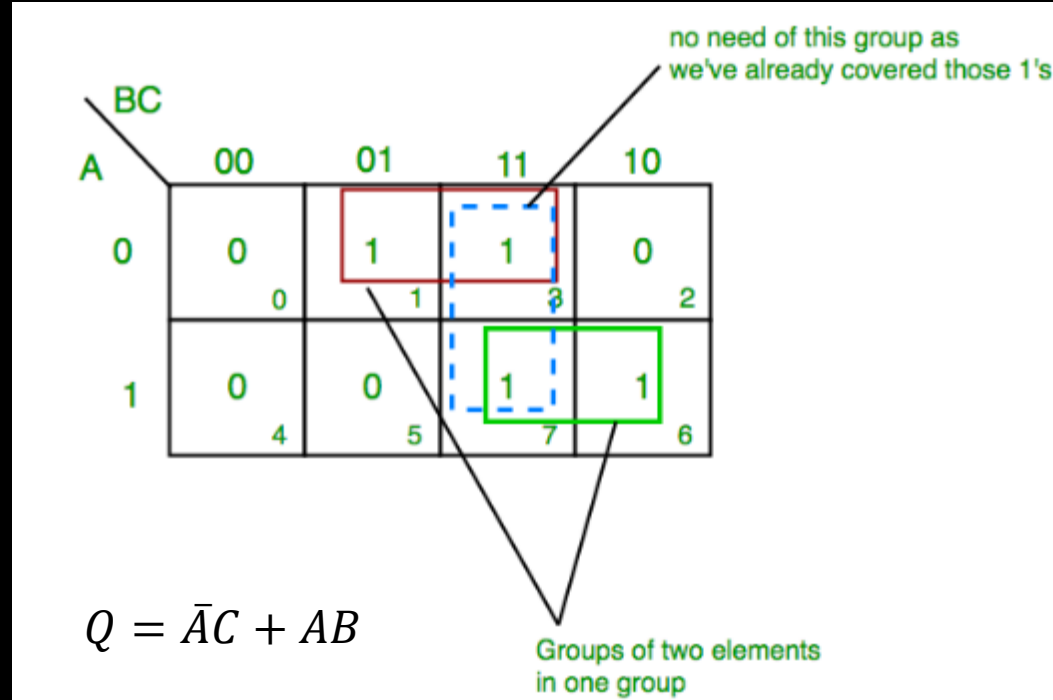
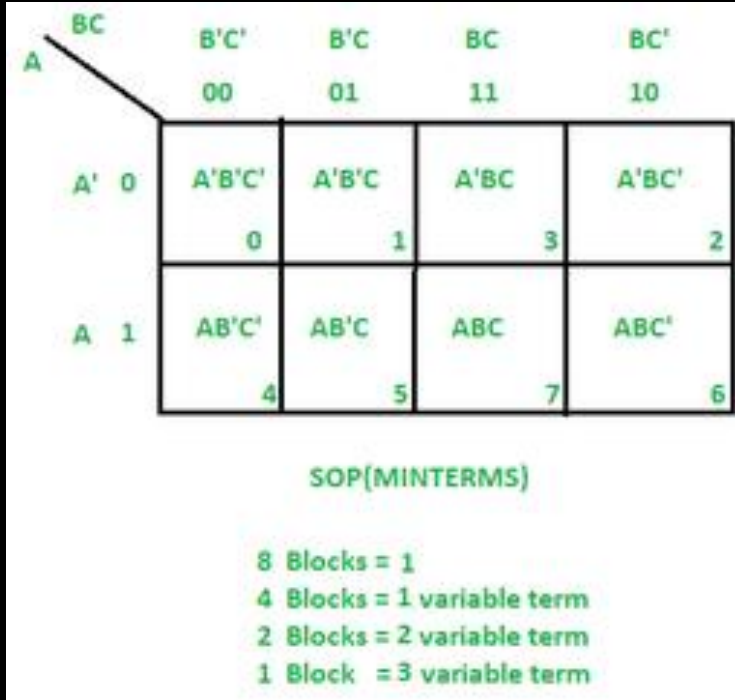


Product of Sum (Max Term)

Inputs			Output	Product
C	B	A	Q	
0	0	0	0	$A+B+C$
0	0	1	1	
0	1	0	0	$A+\bar{B}+C$
0	1	1	1	
1	0	0	1	
1	0	1	0	$\bar{A}+B+\bar{C}$
1	1	0	0	$A+\bar{B}+\bar{C}$
1	1	1	1	



Karnaugh Map (SOP – Min Term)



	A	B	C	Q
0	0	0	0	0
1	0	0	1	1
2	0	1	0	0
3	0	1	1	1
4	1	0	0	0
5	1	0	1	0
6	1	1	0	1
7	1	1	1	1

Karnaugh Map (POS – Max Term)

		BC			
		B+C 00	B+C' 01	B'+C' 11	B'+C 10
A	0	A+B+C 0	A+B+C' 1	A+B'+C' 3	A+B'+C 2
	1	A'+B+C 4	A'+B+C' 5	A'+B'+C' 7	A'+B'+C 6

POS (MAXTERMS)

- 8 Blocks = 0
- 4 Blocks = 1 variable term
- 2 Blocks = 2 variable term
- 1 Block = 3 variable term

		BC			
		00	01	11	10
A	0	0 0	1 1	0 3	1 2
	1	1 4	1 5	0 7	0 6

$$Q = (A + B + C)(\bar{B} + \bar{C})(\bar{A} + \bar{B})$$

	A	B	C	Q
0	0	0	0	0
1	0	0	1	1
2	0	1	0	1
3	0	1	1	0
4	1	0	0	1
5	1	0	1	1
6	1	1	0	0
7	1	1	1	0

Karnaugh Map (POS – Max Term)

		CD			
		C+D	C+D'	C'+D'	C'+D
AB	00	01	11	10	
	A+B	A+B'	A'+B'	A'+B	
00	A+B+C+D 0	A+B+C+D'	A+B+C'+D'	A+B+C'+D 2	
01	A+B'+C+D 4	A+B'+C+D'	A+B'+C'+D'	A+B'+C'+D 6	
11	A'+B'+C+D 12	A'+B'+C+D'	A'+B'+C'+D'	A'+B'+C'+D 14	
10	A'+B+C+D 8	A'+B+C+D'	A'+B+C'+D'	A'+B+C'+D 10	

POS(MAXTERMS)

16 Blocks = 0
 8 Blocks = 1 variable term
 4 Blocks = 2 variable term
 2 Blocks = 3 variable term
 1 Block = 4 variable term

		CD			
		00	01	11	10
AB	00	1 0	1 1	0 3	1 2
	01	1 4	0 5	0 7	1 6
	11	0 12	0 13	1 15	1 14
	10	0 8	1 9	0 11	0 10

	A	B	C	D	Q
0	0	0	0	0	1
1	0	0	0	1	1
2	0	0	1	0	1
3	0	0	1	1	0
4	0	1	0	0	1
5	0	1	0	1	0
6	0	1	1	0	1
7	0	1	1	1	0
8	1	0	0	0	0
9	1	0	0	1	1
10	1	0	1	0	0
11	1	0	1	1	0
12	1	1	0	0	0
13	1	1	0	1	0
14	1	1	1	0	1
15	1	1	1	1	1

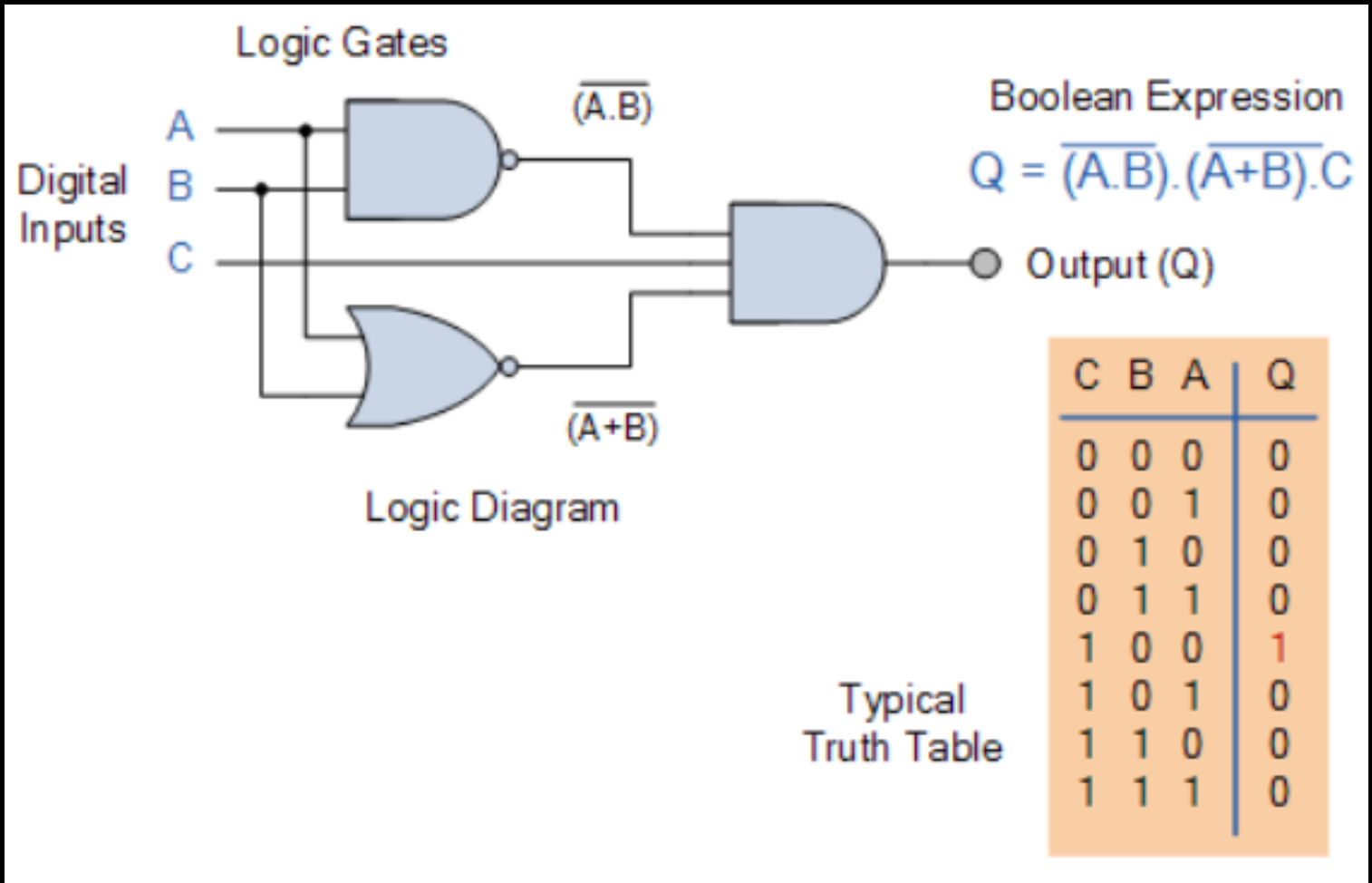
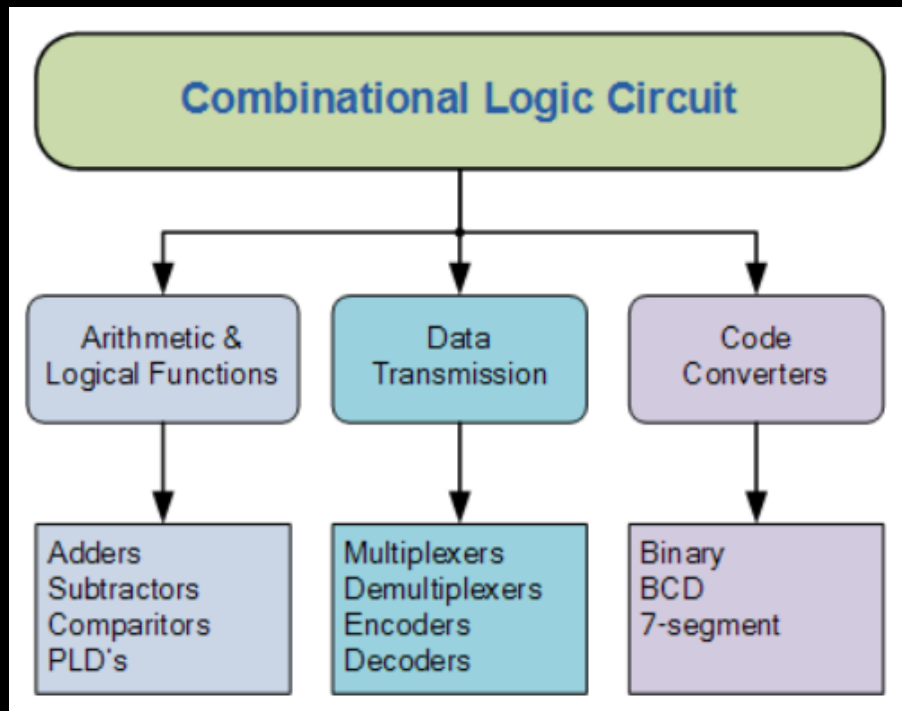
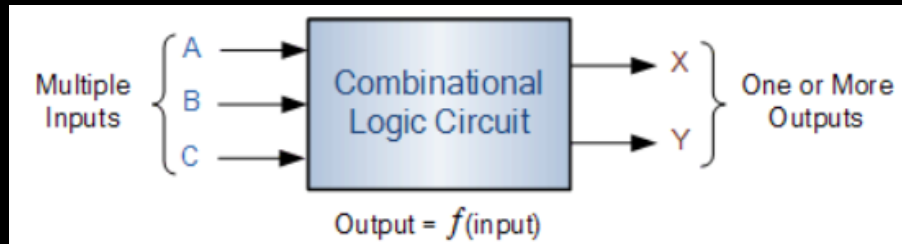
$$Q = (A + \bar{C} + \bar{D})(\bar{B} + C + \bar{D})(\bar{A} + C + D)(\bar{A} + B + \bar{C})$$

Problems

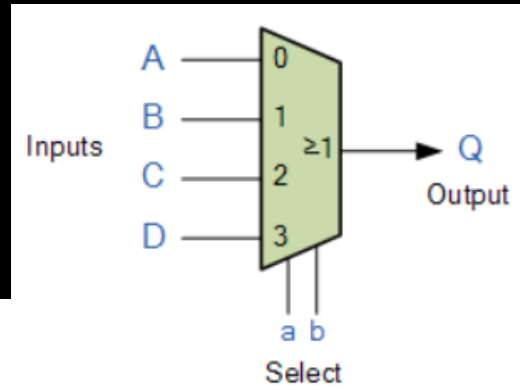
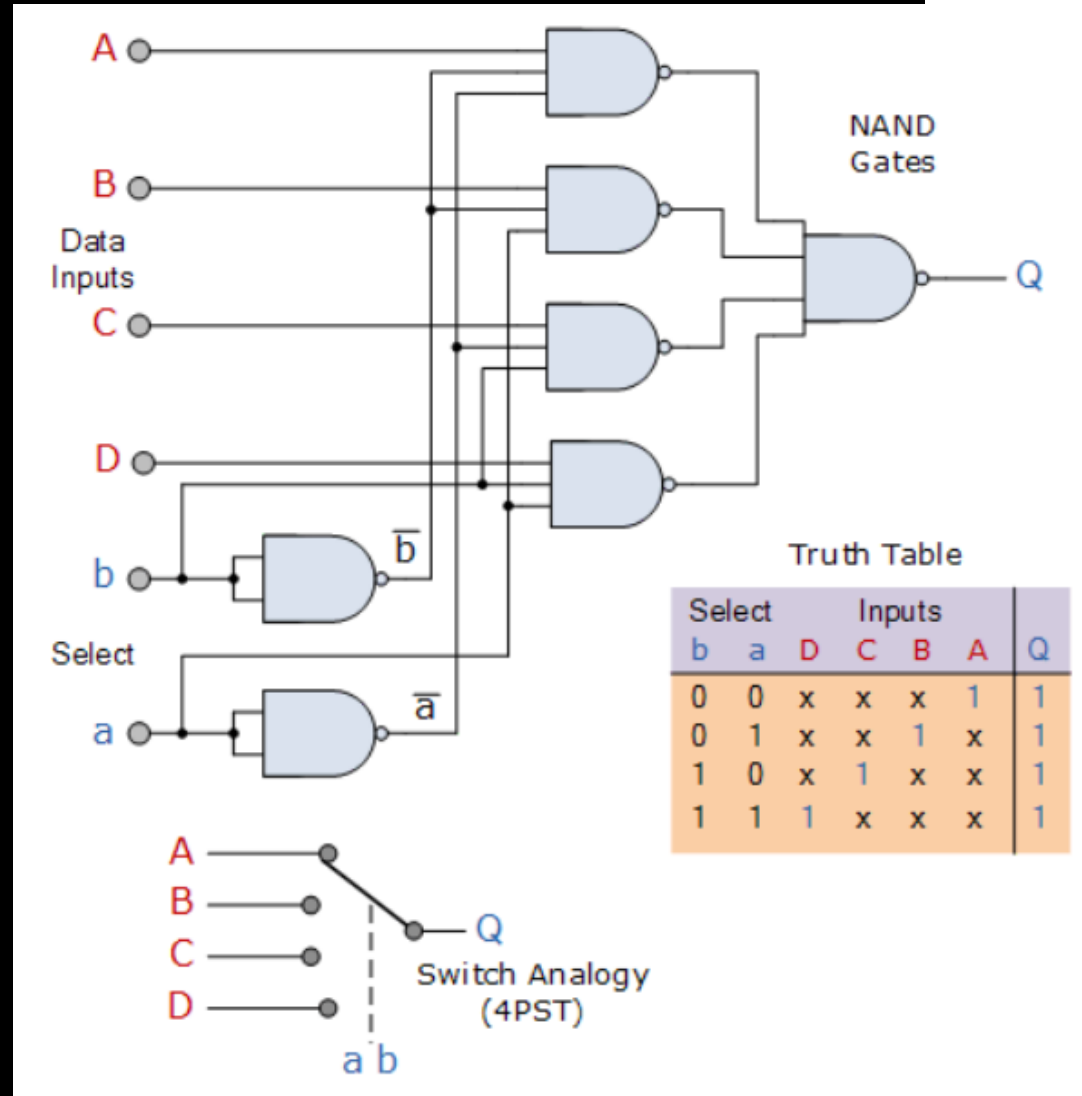
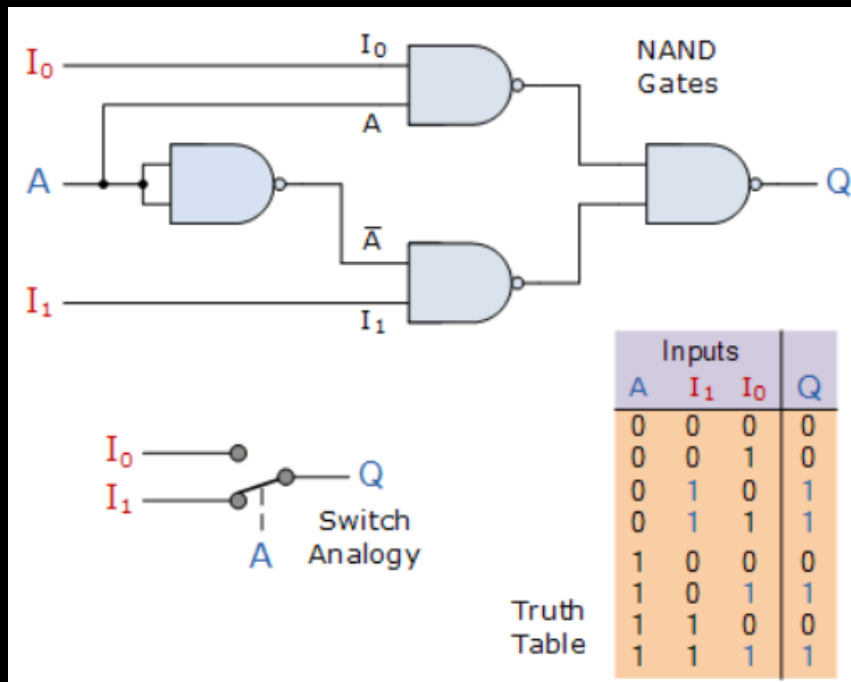
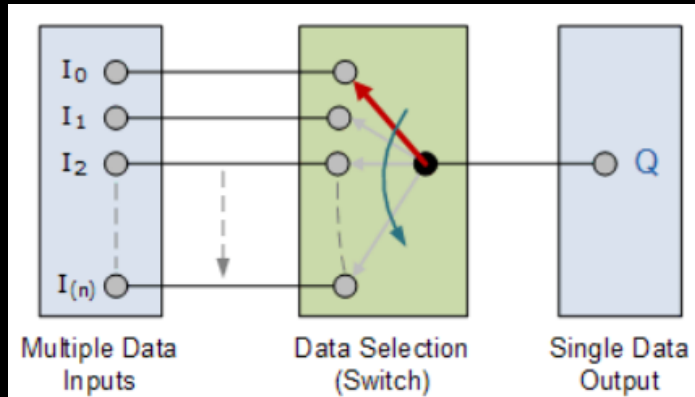
- จาก Truth Table จงใช้ Karnaugh Map เขียน Boolean Algebra ในรูปแบบ Sum of Product และเขียนวงจร Logic Gate ของผลลัพธ์ Q

	A	B	C	D	Q
0	0	0	0	0	1
1	0	0	0	1	1
2	0	0	1	0	1
3	0	0	1	1	0
4	0	1	0	0	1
5	0	1	0	1	0
6	0	1	1	0	1
7	0	1	1	1	0
8	1	0	0	0	0
9	1	0	0	1	1
10	1	0	1	0	0
11	1	0	1	1	0
12	1	1	0	0	0
13	1	1	0	1	0
14	1	1	1	0	1
15	1	1	1	1	1

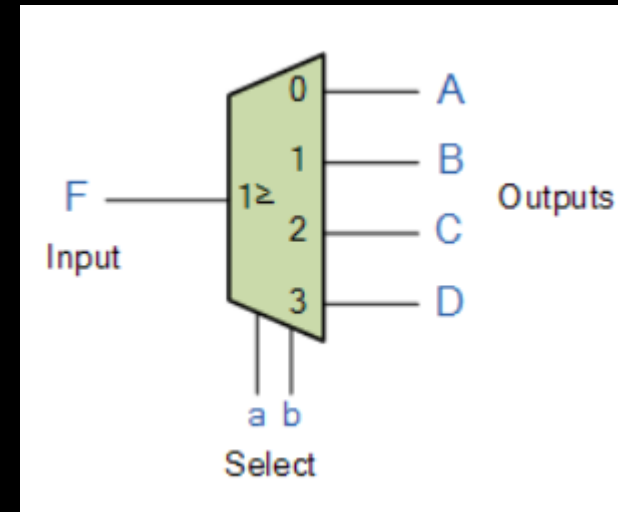
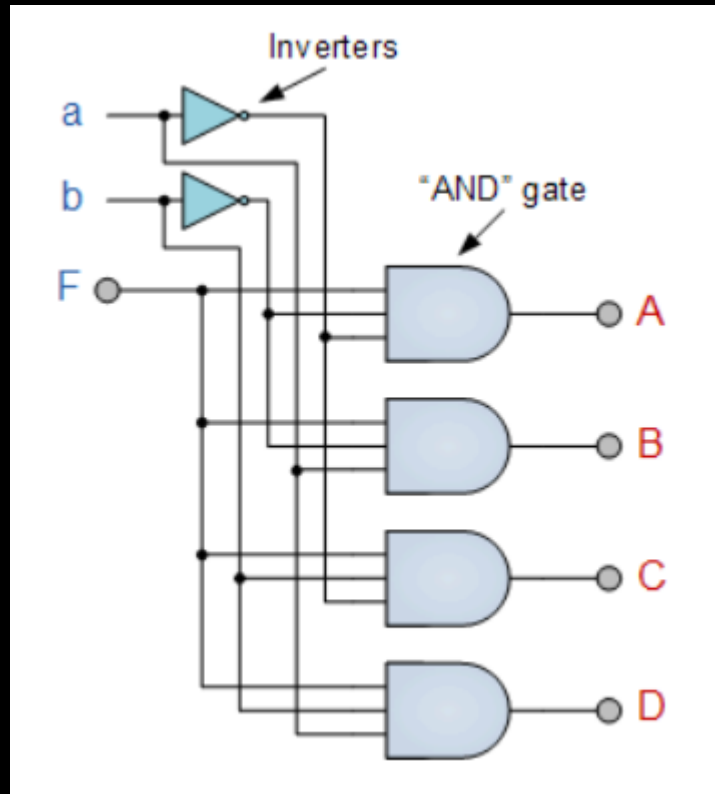
Combinational Circuits



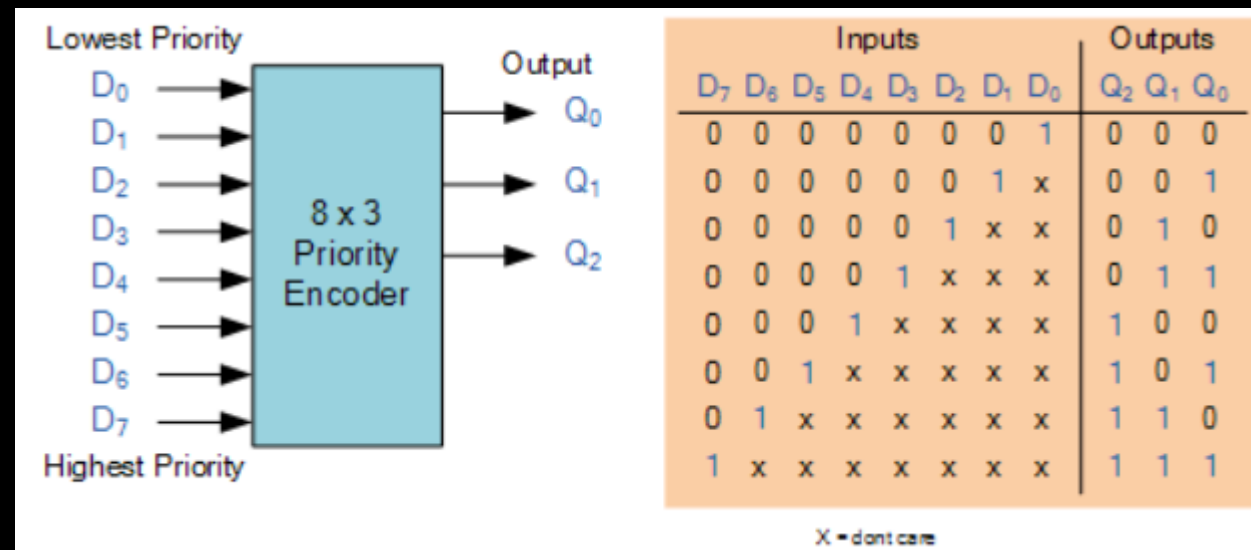
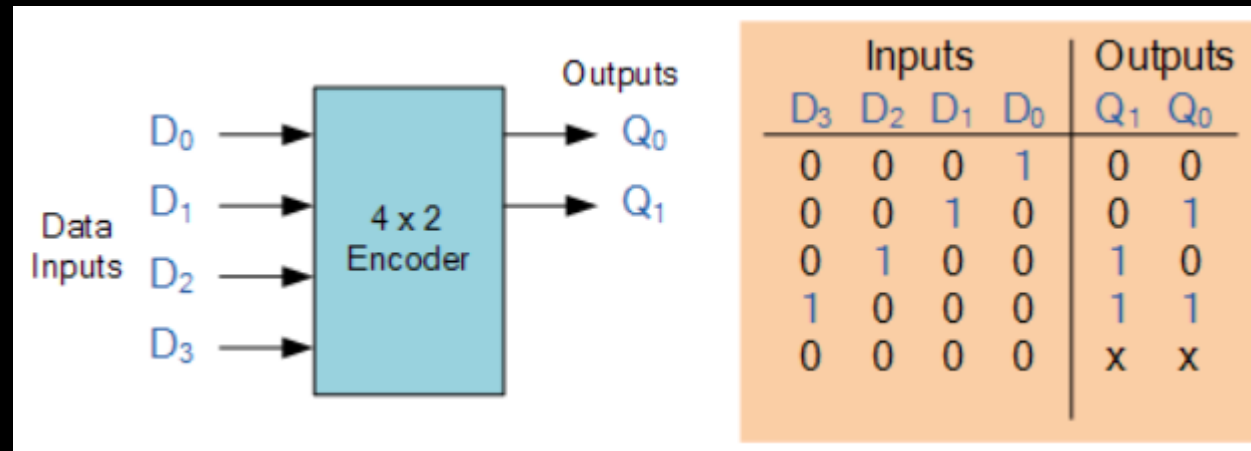
Multiplexer



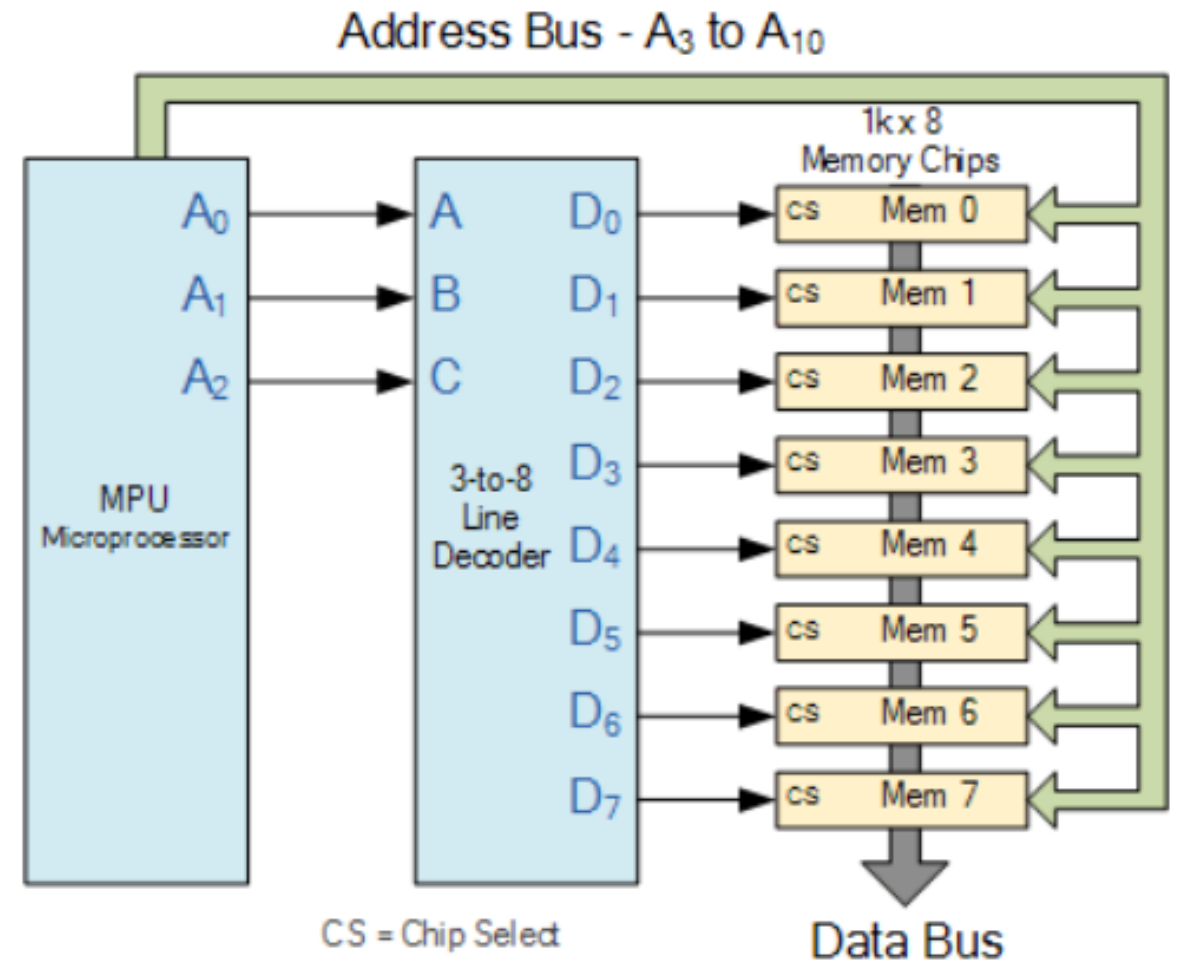
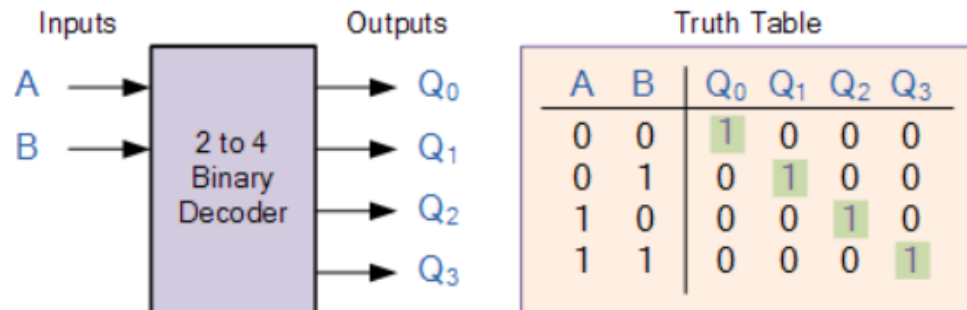
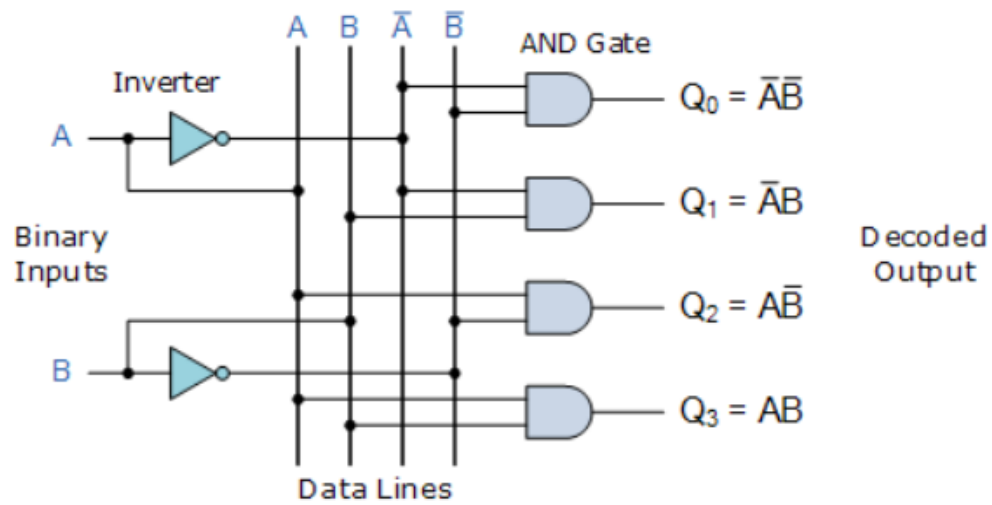
Demultiplexer



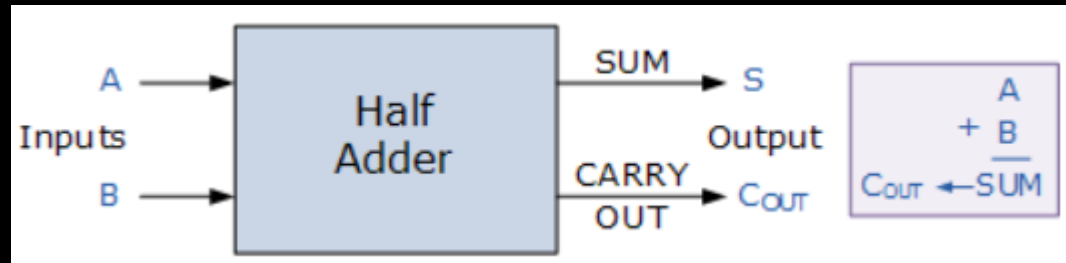
Encoder



Decoder

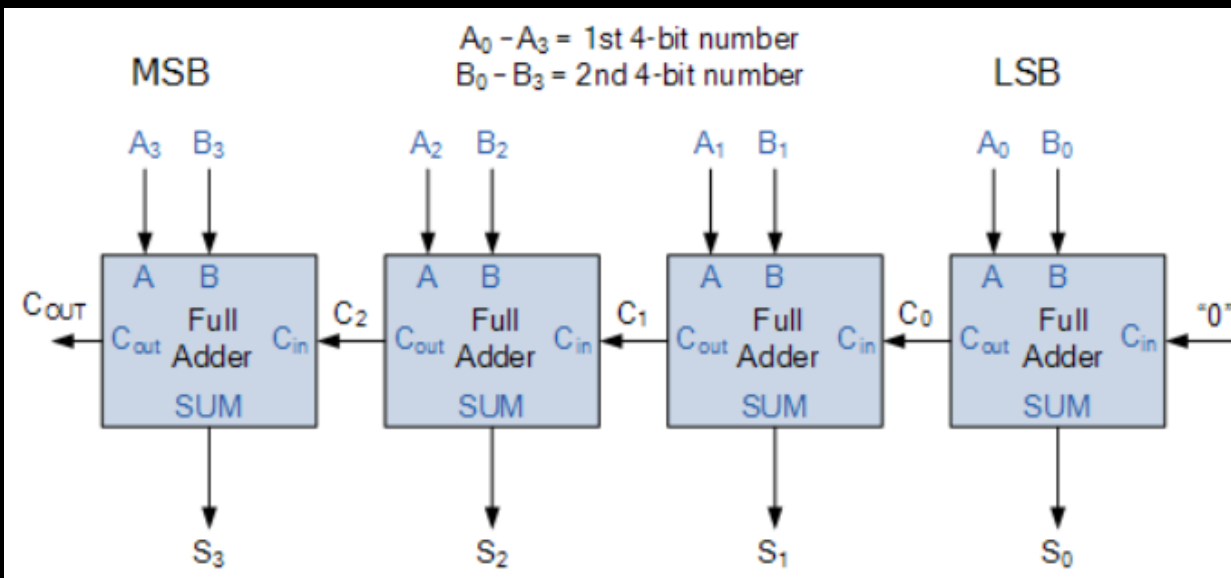
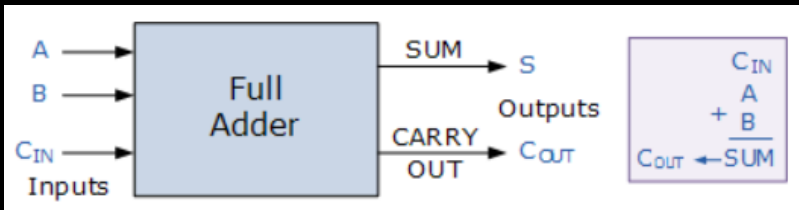


Adders



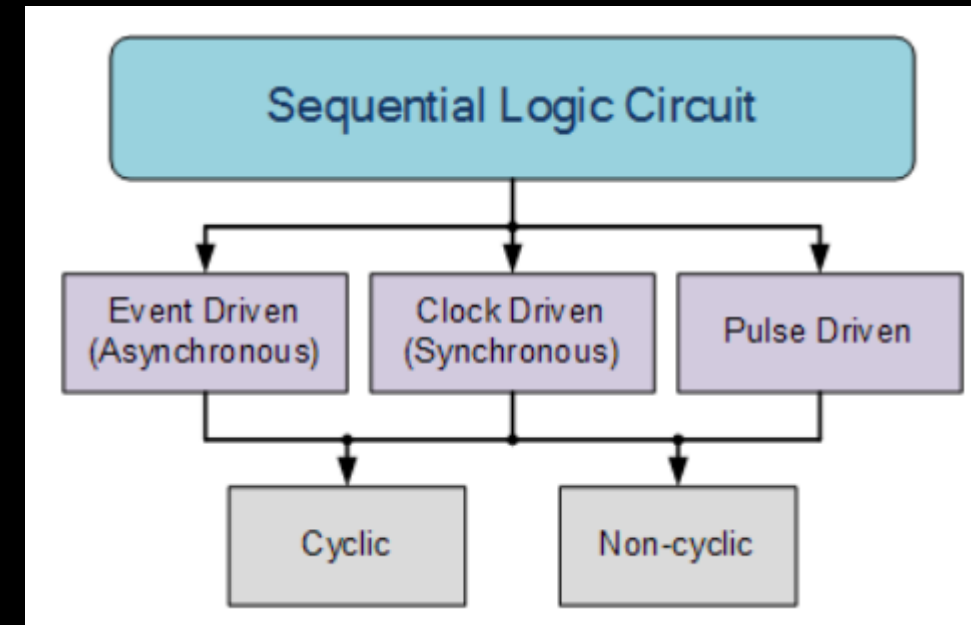
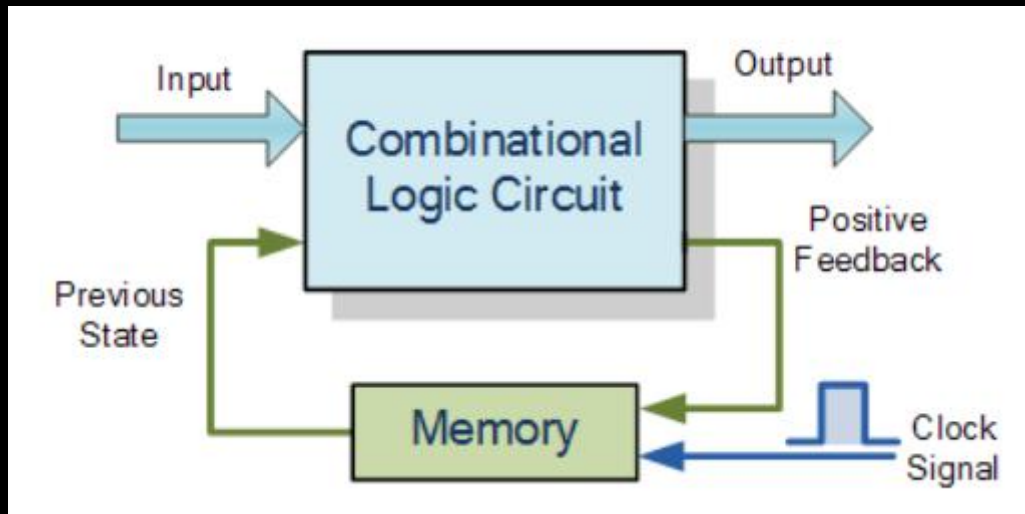
Symbol	Truth Table			
	B	A	SUM	CARRY
	0	0	0	0
	0	1	1	0
	1	0	1	0
	1	1	0	1

Adders



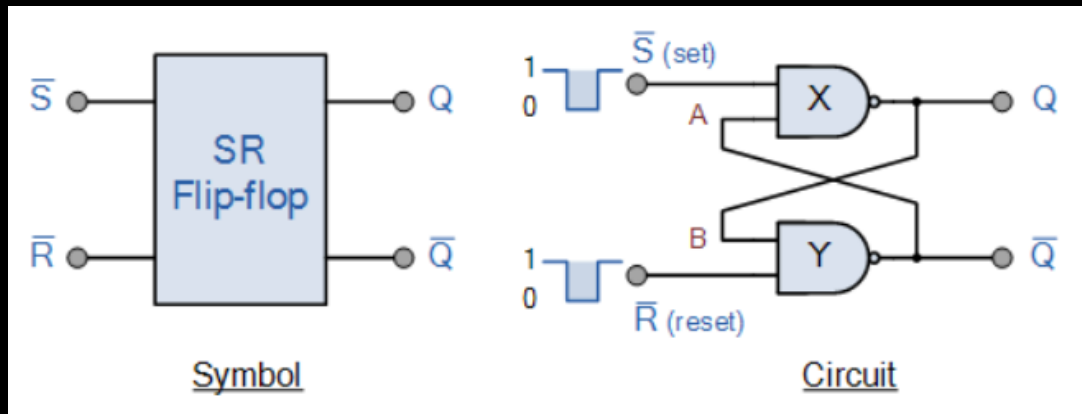
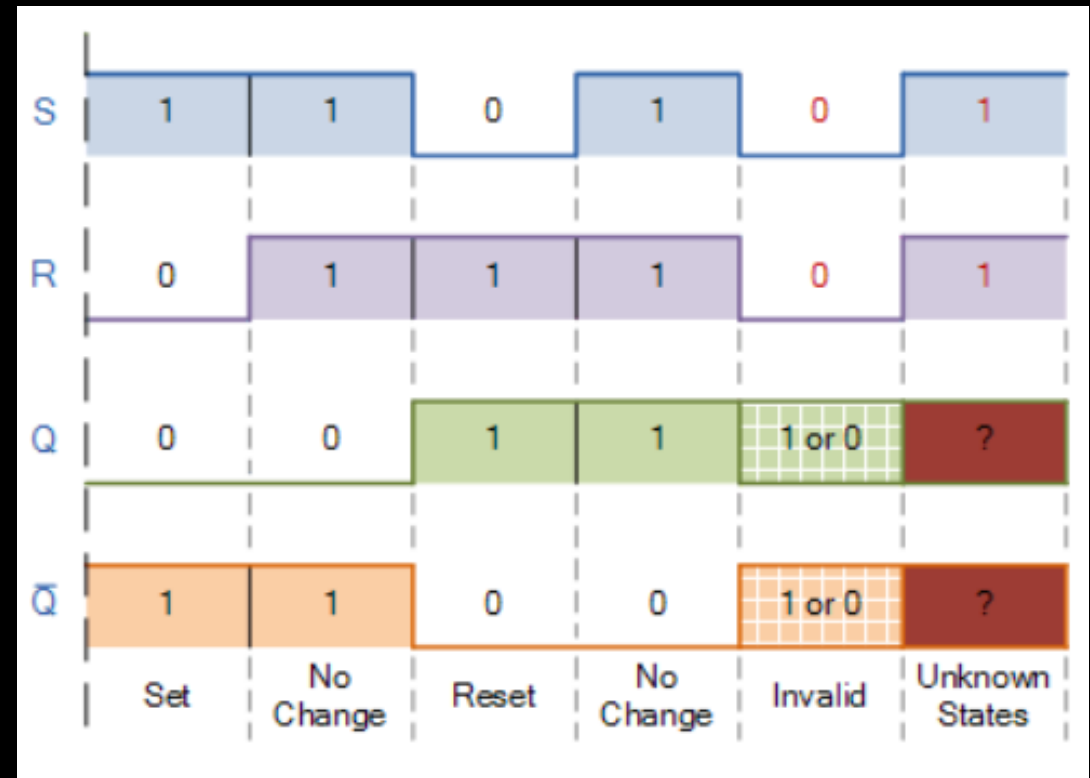
Symbol	Truth Table				
	C-in	B	A	Sum	C-out
	0	0	0	0	0
	0	0	1	1	0
	0	1	0	1	0
	0	1	1	0	1
	1	0	0	1	0
	1	0	1	0	1
	1	1	0	0	1
	1	1	1	1	1

Sequential Circuits

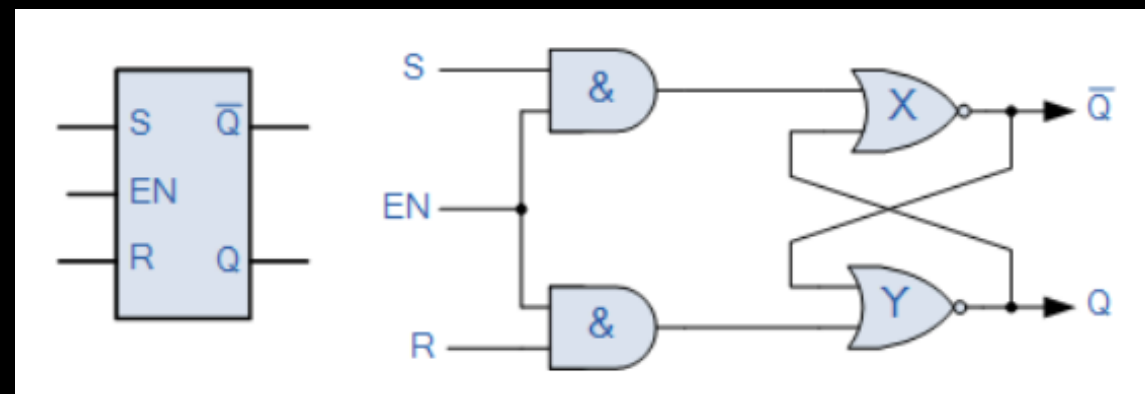
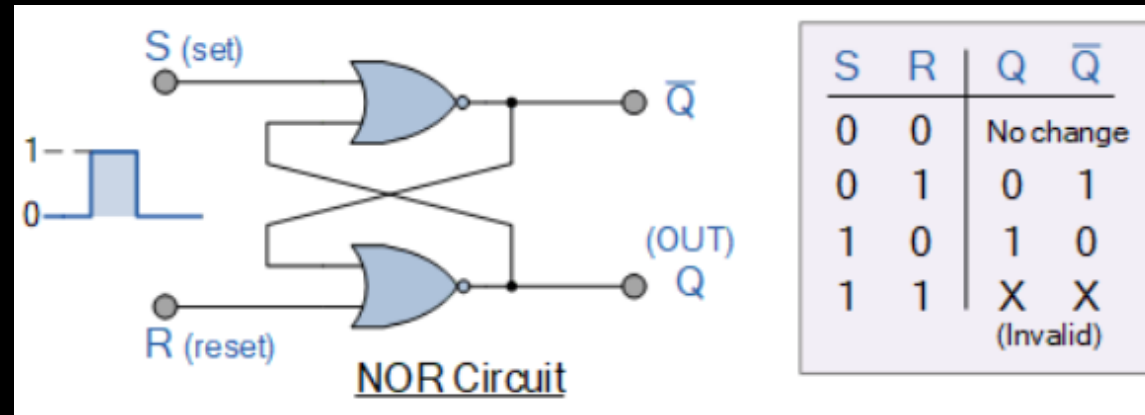


NAND Gate SR Flip-flop

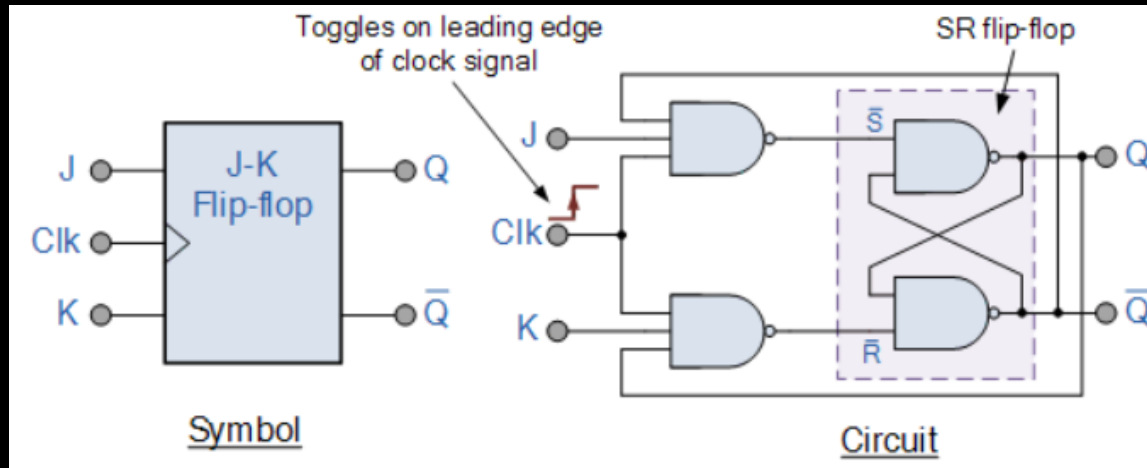
State	S	R	Q	$\bar{Q}$	Description
Set	1	0	0	1	Set $\bar{Q} \gg 1$
	1	1	0	1	no change
Reset	0	1	1	0	Reset $\bar{Q} \gg 0$
	1	1	1	0	no change
Invalid	0	0	1	1	Invalid Condition



NOR Gate SR Flip-flop

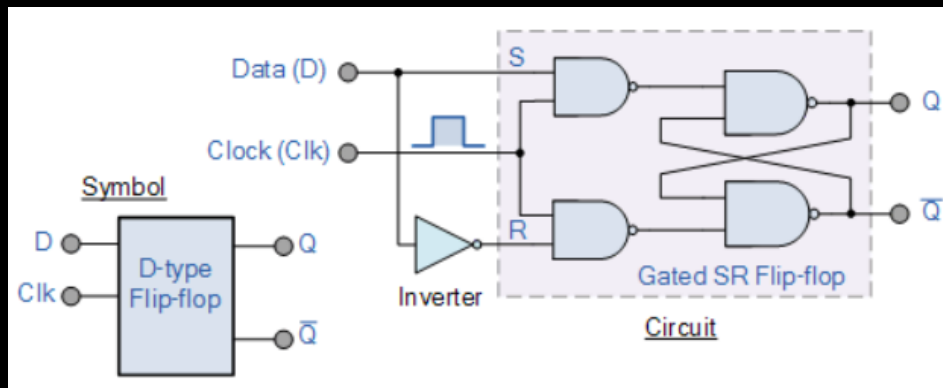


JK Flip-flop

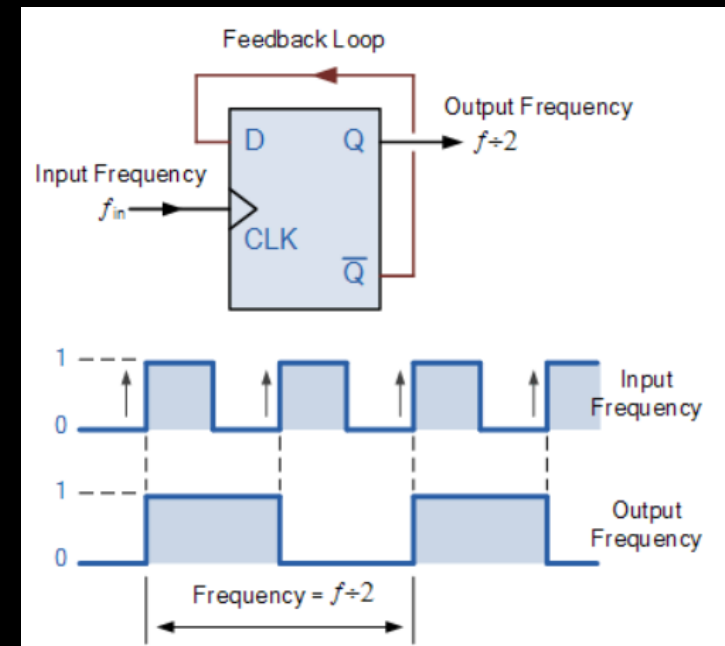
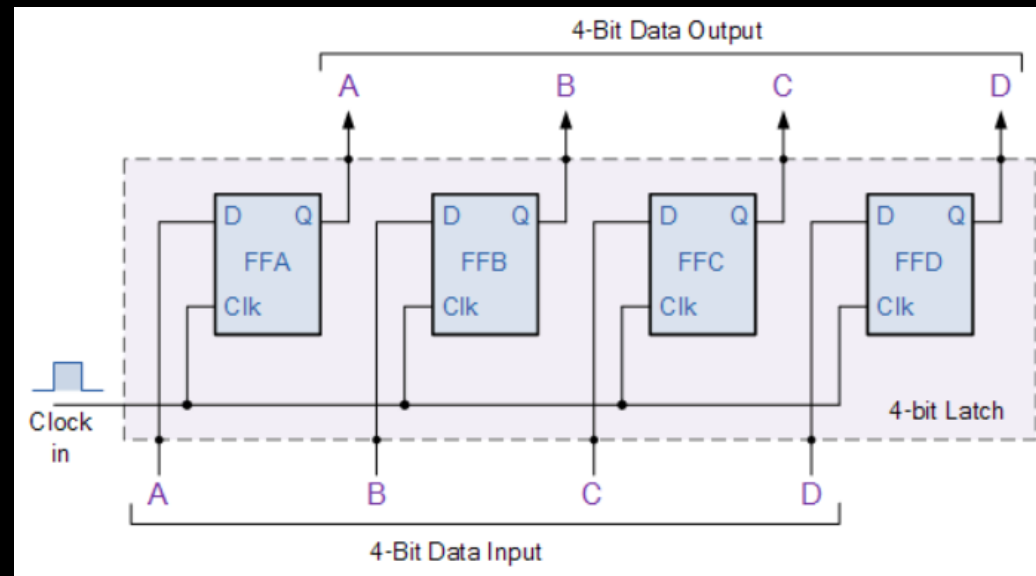


	Clock	Input		Output		Description
	Clk	J	K	Q	$\bar{Q}$	
same as for the SR Latch	X	0	0	1	0	Memory no change
	X	0	0	0	1	
	$\bar{1}$	0	1	1	0	Reset Q » 0
	X	0	1	0	1	
	$\bar{1}$	1	0	0	1	Set Q » 1
	X	1	0	1	0	
toggle action	$\bar{1}$	1	1	0	1	Toggle
	$\bar{1}$	1	1	1	0	

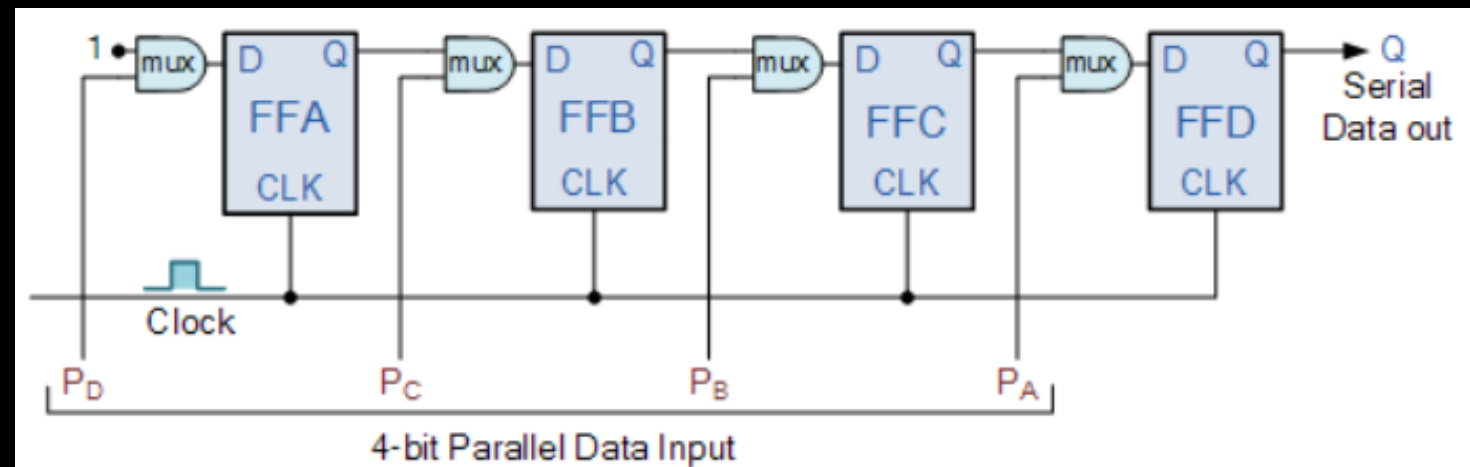
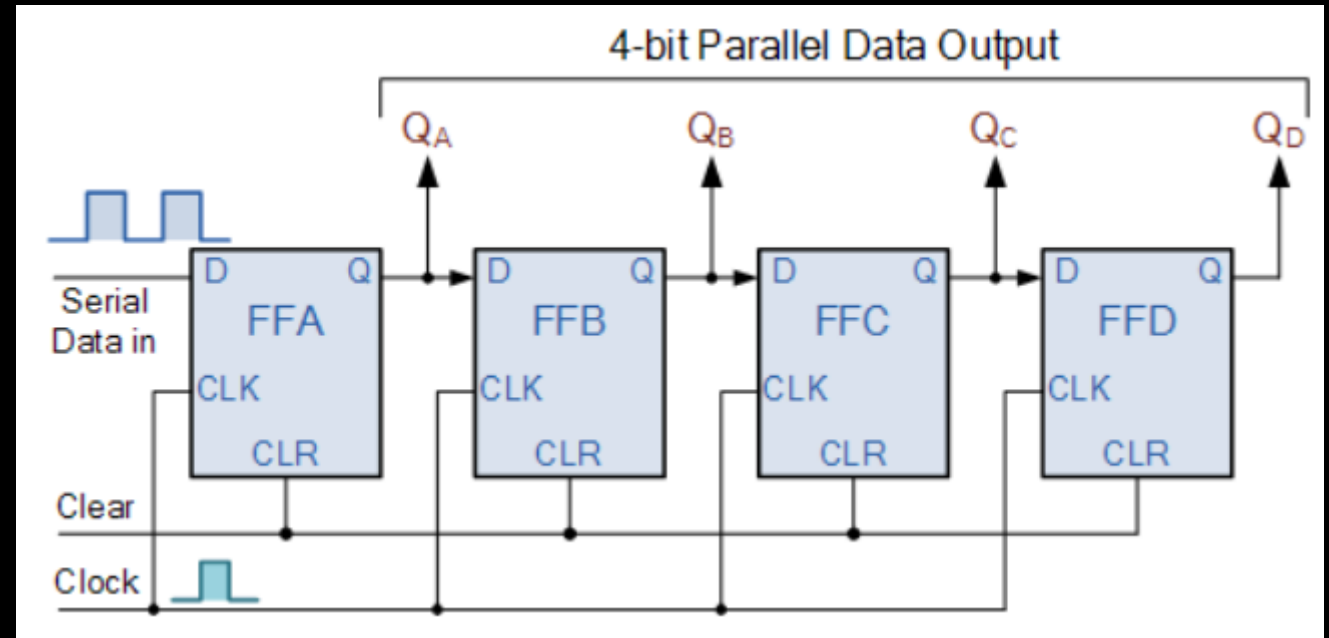
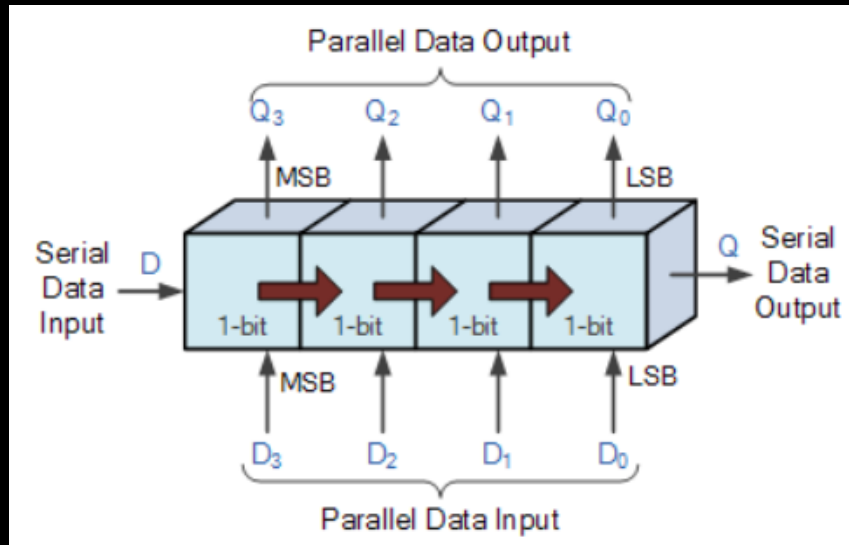
D Flip-flop



Clk	D	Q		Description
$\downarrow \gg 0$	X	Q	$\overline{Q}$	Memory no change
$\uparrow \gg 1$	0	0	1	Reset Q $\gg$ 0
$\uparrow \gg 1$	1	1	0	Set Q $\gg$ 1



Shift Register



Problems

- จงบอกความแตกต่างของวงจร Combinational กับวงจร Sequential

	A	B	C	D	Q
0	0	0	0	0	1
1	0	0	0	1	1
2	0	0	1	0	1
3	0	0	1	1	0
4	0	1	0	0	1
5	0	1	0	1	0
6	0	1	1	0	1
7	0	1	1	1	0
8	1	0	0	0	0
9	1	0	0	1	1
10	1	0	1	0	0
11	1	0	1	1	0
12	1	1	0	0	0
13	1	1	0	1	0
14	1	1	1	0	1
15	1	1	1	1	1